

Linear Transformations IV

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1 Recap on elementary matrices

1.1 Definition

Definition 7.1.1. We identify elementary row matrices and elementary column matrices as follows:

$$E_{i,j}^{\text{I}} = E_{\text{I}}^{i,j}, \quad E_{\lambda,i}^{\text{II}} = E_{\text{II}}^{\lambda,i}, \quad E_{\mu,i,j}^{\text{III}} = E_{\text{III}}^{\mu,i,j}.$$

From now on, we simply refer to them as *elementary matrices*.

1.2 Theorem

Theorem 7.1.1 Every invertible matrix can be written as a product of elementary matrices.

Proof. Let A be an invertible matrix. Since A is invertible,

1. A is an $n \times n$ square matrix;
2. $\text{rank}(A) = n$.

By the QAP Theorem, there exist elementary matrices $E_1, E_2, \dots, E_k$, $G_1, G_2, \dots, G_\ell$, s.t. $(E_1 E_2 \cdots E_k) A (G_1 G_2 \cdots G_\ell) = \underbrace{I_n}_{\text{SNF of } A}$.

$$\begin{aligned} \text{Hence, } A &= (E_1 E_2 \cdots E_k)^{-1} I_n (G_1 G_2 \cdots G_\ell)^{-1}, \\ \implies A &= E_k^{-1} \cdots E_2^{-1} E_1^{-1} G_\ell^{-1} \cdots G_2^{-1} G_1^{-1}. \end{aligned}$$

Since the inverse of every elementary matrix is again an elementary matrix, A is a product of elementary matrices. $\square$

2 Augmented matrices

Definition

Definition 7.2.1. Let $A \in M_{m \times p}(F)$, $B \in M_{m \times q}(F)$. The *augmented matrix* $(A \mid B)$ is an $m \times (p + q)$ matrix for which the first p columns give the matrix A and the last q columns give the matrix B .

Note. The vertical line is placed to separate the matrices A and B for convenience only. Let

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}.$$

Then

$$(A \mid I_2) = \left(\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 3 & 4 & 0 & 1 \end{array} \right).$$

3 Application: Finding inverse of a matrix

Let A be an $(n \times n)$ invertible matrix. Consider the augmented matrix $(A \mid I_n)$ formed by A and I_n . The key observation is that

$$A^{-1}(A \mid I_n) = (I_n \mid A^{-1}).$$

By Theorem 7.1.1, A^{-1} , as an invertible matrix, can be written as a product of elementary matrices. Therefore,

$$\underbrace{(E_1 E_2 \cdots E_k)}_{A^{-1}}(A \mid I_n) = (I_n \mid A^{-1}),$$

where

$$A^{-1} = E_1 E_2 \cdots E_k.$$

Since multiplying elementary matrices from the left corresponds to conducting *row operations*, the above operations give us a concrete strategy to compute the inverse of a given matrix A , which we record below.

3.1 Theorem

Theorem 7.3.1. Let A be an $(n \times n)$ invertible matrix. If we conduct row operations to transform the augmented matrix

$$(A \mid I_n) \xrightarrow{\text{row operations}} (I_n \mid B),$$

Then $B = A^{-1}$.

3.2 Example

3.2.1 Example 1

Compute the inverse of the matrix

$$A = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 3 \\ 2 & 4 & -1 \end{pmatrix}.$$

Solution. We aim to transform the augmented matrix $(A \mid I_3)$ into the form $(I_3 \mid B)$ by row operations.

$$\begin{aligned}
(A \mid I_3) &= \left(\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 0 & 1 & 3 & 0 & 1 & 0 \\ 2 & 4 & -1 & 0 & 0 & 1 \end{array} \right) \\
&\longrightarrow \left(\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 0 & 1 & 3 & 0 & 1 & 0 \\ 0 & 0 & -3 & -2 & 0 & 1 \end{array} \right) \\
&\longrightarrow \left(\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 0 & 1 & 3 & 0 & 1 & 0 \\ 0 & 0 & 1 & \frac{2}{3} & 0 & -\frac{1}{3} \end{array} \right) \\
&\longrightarrow \left(\begin{array}{ccc|ccc} 1 & 2 & 0 & \frac{1}{3} & 0 & \frac{1}{3} \\ 0 & 1 & 0 & -2 & 1 & 1 \\ 0 & 0 & 1 & \frac{2}{3} & 0 & -\frac{1}{3} \end{array} \right) \\
&\longrightarrow \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{13}{3} & -2 & -\frac{5}{3} \\ 0 & 1 & 0 & -2 & 1 & 1 \\ 0 & 0 & 1 & \frac{2}{3} & 0 & -\frac{1}{3} \end{array} \right).
\end{aligned}$$

Thus,

$$A^{-1} = \begin{pmatrix} \frac{13}{3} & -2 & -\frac{5}{3} \\ -2 & 1 & 1 \\ \frac{2}{3} & 0 & -\frac{1}{3} \end{pmatrix}.$$

□

3.2.2 Example 2

Consider the function $T : \mathbb{R}[x]_{\leq 2} \rightarrow \mathbb{R}[x]_{\leq 2}$ defined by

$$T(p(x)) = p(x) + p'(x) + p''(x).$$

Let $B = \{1, x, x^2\}$ be a basis of $\mathbb{R}[x]_{\leq 2}$.

- (a) Prove that T is a linear transformation.
- (b) Find $[T]_B^B$. Is T an isomorphism?
- (c) Write down the inverse function T^{-1} .

Solution. (a) For any $p(x), q(x) \in \mathbb{R}[x]_{\leq 2}$,

$$\begin{aligned} T(p(x) + q(x)) &= (p(x) + q(x)) + (p(x) + q(x))' + (p(x) + q(x))'' \\ &= p(x) + q(x) + p'(x) + q'(x) + p''(x) + q''(x) \\ &= (p(x) + p'(x) + p''(x)) + (q(x) + q'(x) + q''(x)) \\ &= T(p(x)) + T(q(x)). \end{aligned}$$

Also, for any $\alpha \in \mathbb{R}$,

$$\begin{aligned} T(\alpha p(x)) &= \alpha p(x) + (\alpha p(x))' + (\alpha p(x))'' \\ &= \alpha p(x) + \alpha p'(x) + \alpha p''(x) \\ &= \alpha(p(x) + p'(x) + p''(x)) \\ &= \alpha T(p(x)). \end{aligned}$$

Therefore, T is a linear transformation.

(b) We have

$$\begin{aligned} T(\textcolor{red}{1}) &= 1 = 1 \cdot 1 + 0 \cdot x + 0 \cdot x^2, \\ T(\textcolor{red}{x}) &= x + 1 = 1 \cdot 1 + 1 \cdot x + 0 \cdot x^2, \\ T(\textcolor{red}{x}^2) &= x^2 + 2x + 2 = 2 \cdot 1 + 2 \cdot x + 1 \cdot x^2. \end{aligned}$$

So, $[T]_B^B = \begin{pmatrix} 1 & 1 & 2 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix}$ is already in REF, and

$$\text{rank}[T]_B^B = 3 = \dim(\mathbb{R}[x]_{\leq 2}) \implies \text{surjective}$$

$$\text{nullity}[T]_B^B = \underbrace{\textcolor{red}{3}}_{\text{dim of domain}} - \underbrace{3}_{\text{rank}} = 0 \implies \text{injective}$$

so T is an isomorphism.

(c) We first compute the inverse matrix. Note: $[T^{-1}]_B^B = ([T]_B^B)^{-1}$

Consider

$$\left(\begin{array}{ccc|ccc} 1 & 1 & 2 & 1 & 0 & 0 \\ 0 & 1 & 2 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right) \longrightarrow \left(\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & -2 \\ 0 & 1 & 0 & 0 & 1 & -2 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right)$$

and then

$$\longrightarrow \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & -1 & 0 \\ 0 & 1 & 0 & 0 & 1 & -2 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right).$$

Therefore,

$$[T^{-1}]_B^B = ([T]_B^B)^{-1} = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{pmatrix}.$$

For

$$p(x) = a + bx + cx^2,$$

we have

$$\begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} a - b \\ b - 2c \\ c \end{pmatrix}.$$

Translating back to $\mathbb{R}[x]_{\leq 2}$,

$$\boxed{T^{-1}(a + bx + cx^2) = (a - b) + (b - 2c)x + cx^2}.$$

□

4 Application: System of linear equations

Historically, matrices arise when one attempts to solve systems of linear equations. For example, consider the system

$$\begin{cases} x + 2y = 3, \\ 3x + 4y = 1. \end{cases}$$

A condensed way to write the above system is $\begin{pmatrix} 1 & 2 & 3 \\ 3 & 4 & 1 \end{pmatrix}$. This is called the *augmented matrix* for the above system of equations.

4.1 Definition

Definition 7.4.1. Let $(A \mid \mathbf{b})$ be the augmented matrix associated with a system of linear equations $A\mathbf{x} = \mathbf{b}$. We call the system of equations

- *homogeneous* if $\mathbf{b} = \mathbf{0}$;
- *inhomogeneous* if $\mathbf{b} \neq \mathbf{0}$.

4.2 Theorem

Theorem 7.4.1. The solution set of a homogeneous system $(A \mid \mathbf{0})$ coincides with the subspace $\ker(A)$. In particular,

$$\underbrace{\dim(\ker(A))}_{\text{nullity}(A)} = (\text{number of columns of } A) - \text{rank}(A).$$

Proof. A vector $\mathbf{x}$ is a solution of $(A \mid \mathbf{0}) \Leftrightarrow A\mathbf{x} = \mathbf{0}$

$$\implies (A \mid \mathbf{0}) \Leftrightarrow \mathbf{x} \in \ker(A).$$

Moreover, by the rank-nullity theorem,

$$\dim(\ker(A)) = \text{nullity}(A) = (\text{number of columns of } A) - \text{rank}(A).$$

□

Theorem 7.4.2. Let $A \in M_{m \times n}(F)$. If $m < n$, then the homogeneous system $(A \mid \mathbf{0})$ has a non-zero solution.

Proof. Since $A \in M_{m \times n}(F)$ represents a linear map $F^n \rightarrow F^m$, we have $\text{im}(A) \subseteq F^m$. Therefore, $\text{rank}(A) \leq m$. By the rank-nullity theorem,

$$\begin{aligned} \dim(\ker(A)) &= (\text{number of columns of } A) - \text{rank}(A) \\ &= n - \text{rank}(A) \\ &\geq n - m \geq 1 \implies n \geq m \end{aligned}$$

Hence, $\ker(A) \neq \{\mathbf{0}\}$. Therefore, there $\exists$ some non-zero vector $\mathbf{x} \in \ker(A)$ s.t. $A\mathbf{x} = \mathbf{0}$.

Thus, the homogeneous system $(A \mid \mathbf{0})$ has a non-zero solution. □

Theorem 7.4.3. Let $(A \mid \mathbf{b})$ be the augmented matrix of a general system of equations. Then the system is *consistent* (has a solution) iff.

$$\text{rank}(A) = \text{rank}(A \mid \mathbf{b}).$$

Proof. Let $A = (\mathbf{c}_1 \ \mathbf{c}_2 \ \cdots \ \mathbf{c}_n)$.

($\Rightarrow$)

Suppose $\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$ is a solution of $A\mathbf{x} = \mathbf{b}$.

$$\implies x_1\mathbf{c}_1 + x_2\mathbf{c}_2 + \cdots + x_n\mathbf{c}_n = \mathbf{b}.$$

$$\implies \mathbf{b} \in \text{span}\{\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_n\}.$$

$$\implies \text{span}\{\underbrace{\mathbf{c}_1, \dots, \mathbf{c}_n}_{\text{columns of } A}\} = \text{span}\{\underbrace{\mathbf{c}_1, \dots, \mathbf{c}_n, \mathbf{b}}_{\text{columns of } (A \mid \mathbf{b})}\}.$$

$$\implies \text{im}(\mathbf{A}) = \text{im}(\mathbf{A} \mid \mathbf{b}),$$

$$\implies \text{rank}(A) = \text{rank}(A \mid \mathbf{b}).$$

($\Leftarrow$) Conversely, suppose $\text{rank}(A) = \text{rank}(A \mid \mathbf{b})$.

$$\implies \text{im}(A) \subseteq \text{im}(A \mid \mathbf{b}).$$

Since these two subspaces have the same dimension, they must be equal

$$\implies \text{im}(A) = \text{im}(A \mid \mathbf{b}).$$

Thus, $\mathbf{b} \in \text{span}\{\mathbf{c}_1, \dots, \mathbf{c}_n\}$.

$$\implies \exists x_1, \dots, x_n \text{ s.t. } x_1 \mathbf{c}_1 + \dots + x_n \mathbf{c}_n = \mathbf{b}. \text{ Equivalently, there } \exists \mathbf{x} \text{ s.t. } A\mathbf{x} = \mathbf{b}.$$

Therefore, the system is consistent. □

Theorem 7.4.4. (Principle of linearity) Suppose $\mathbf{v}$ is a solution to the system of equations $(A \mid \mathbf{b})$. Then the general solution of $(A \mid \mathbf{b})$ is given by

$$\mathbf{v} + \ker(A) := \{\mathbf{v} + \mathbf{n} : \mathbf{n} \in \ker(A)\}.$$

Proof. Given that $\mathbf{v}$ is a solution of $(A \mid \mathbf{b})$, we have $A\mathbf{v} = \mathbf{b}$ (1)

Suppose $\mathbf{u}$ is another solution of $(A \mid \mathbf{b})$. Then $A\mathbf{u} = \mathbf{b}$ (2)

Consider (2) – (1):

$$\begin{aligned} A\mathbf{u} - A\mathbf{v} &= \mathbf{b} - \mathbf{b} \\ \iff A(\mathbf{u} - \mathbf{v}) &= \mathbf{0} \\ \iff \mathbf{u} - \mathbf{v} &\in \ker(A) \\ \iff \mathbf{u} - \mathbf{v} = \mathbf{n} &\text{ for some } \mathbf{n} \in \ker(A) \\ \iff \mathbf{u} = \mathbf{v} + \mathbf{n} &\text{ for some } \mathbf{n} \in \ker(A). \end{aligned}$$

Hence, the general solution of $(A \mid \mathbf{b})$ is $\mathbf{v} + \ker(A)$. □

Remark. In particular, the system $(A \mid \mathbf{b})$ has a **unique** solution iff.

$$\begin{cases} \text{rank}(A) = \text{rank}(A \mid \mathbf{b}), \\ \ker(A) = \{\mathbf{0}\}. \end{cases}$$

□

4.3 Example

4.3.1 General systems: Spanning

Does $(3, 3, 2)$ belong to $\text{span}\{(1, 1, 1), (1, -1, 0)\}$?

Solution. Suppose it does. Then there $\exists a, b \in \mathbb{R}$ s.t.

$$(3, 3, 2) = a(1, 1, 1) + b(1, -1, 0).$$

Equivalently,

$$\begin{cases} a + b = 3, \\ a - b = 3, \\ a = 2. \end{cases}$$

Consider the coefficient matrix

$$A = \begin{pmatrix} 1 & 1 \\ 1 & -1 \\ 1 & 0 \end{pmatrix}.$$

We have $\text{rank}(A) = 2$.

On the other hand, the augmented matrix is

$$(A \mid \mathbf{b}) = \left(\begin{array}{cc|c} 1 & 1 & 3 \\ 1 & -1 & 3 \\ 1 & 0 & 2 \end{array} \right),$$

and $\text{rank}(A \mid \mathbf{b}) = 3$.

Thus, $\text{rank}(A) \neq \text{rank}(A \mid \mathbf{b})$, so the corresponding system is inconsistent.

Hence, $\boxed{(3, 3, 2) \notin \text{span}\{(1, 1, 1), (1, -1, 0)\}}$.

□

4.3.2 General systems: Linear Transformations

Consider the linear transformation $T : \mathbb{R}[x]_{\leq 2} \rightarrow \mathbb{R}[x]_{\leq 2}$ defined by

$$T(a + bx + cx^2) = (a + b + c) + (a - b + c)x + (a + c)x^2.$$

Does $3 + 3x + 2x^2$ belong to $\text{im}(T)$?

Solution. Suppose it does. $\implies 3 + 3x + 2x^2 = T(a + bx + cx^2)$ for some $a, b, c \in \mathbb{R}$,

By comparing the coefficients, we obtain

$$\begin{cases} a + b + c = 3, \\ a - b + c = 3, \\ a + c = 2. \end{cases} \quad (*)$$

The coefficient matrix of $(*)$ is

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 0 & 1 \end{pmatrix},$$

and $\text{rank}(A) = 2$.

On the other hand,

$$\text{rank} \left(\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 1 & -1 & 1 & 3 \\ 1 & 0 & 1 & 2 \end{array} \right) = 3.$$

Therefore, $\text{rank}(A) \neq \text{rank}(A \mid \mathbf{b})$, so $(*)$ has no solution.

Hence, $\boxed{3 + 3x + 2x^2 \notin \text{im}(T)}$.

□

4.3.3 Inhomogeneous systems

Consider the inhomogeneous system $\begin{cases} x_1 + 2x_2 + 3x_3 = 4, \\ 5x_1 + 6x_2 + 7x_3 = 8. \end{cases}$ Find the general solution.

Solution. The general solution is given by

$$\left\{ \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} -2 + t \\ 3 - 2t \\ t \end{pmatrix} : t \in \mathbb{R} \right\}.$$

$$= \left\{ \begin{pmatrix} -2 \\ 3 \\ 0 \end{pmatrix} + t \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} : t \in \mathbb{R} \right\}$$

$$= \begin{pmatrix} -2 \\ 3 \\ 0 \end{pmatrix} + \text{span} \left\{ \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} \right\}.$$

Here, $\begin{pmatrix} -2 \\ 3 \\ 0 \end{pmatrix}$ is a particular solution, while $\text{span} \left\{ \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} \right\} = \ker \begin{pmatrix} 1 & 2 & 3 \\ 5 & 6 & 7 \end{pmatrix}$.

Thus, the general solution has the form $\boxed{\text{particular solution} + \ker(A)}$. □

5 TFAE Theorem Ver.1

5.1 Theorem

Theorem 7.5.1. Let $T : V \rightarrow W$ be a linear transformation s.t.

$$\dim(V) = \dim(W) < \infty.$$

Therefore, having fixed bases, T can be represented by a **square matrix** A .

The following are equivalent (TFAE).

1. T is injective.
2. $\ker(T) = \{\mathbf{0}_V\}$.
3. $\text{nullity}(T) = 0$.
4. $\text{rank}(T) = \dim(W)$ (T has full rank).
5. $\text{im}(T) = W$.
6. T is surjective.
7. T is an isomorphism (bijective linear map).
8. A is invertible (A^{-1} exists).
9. $\ker(A) = \{\mathbf{0}\}$.
10. A has full rank.
11. All the rows of A are linearly independent.
12. All the columns of A are linearly independent.
13. The system of equations $A\mathbf{x} = \mathbf{b}$ has a unique solution, namely $\mathbf{x} = A^{-1}\mathbf{b}$.

5.2 Example

5.2.1 Prove that a matrix is invertible

Let $A \in M_n(F)$. It is known that $A^k = \mathbf{0}$ for some positive integer k .
Prove that $A + I_n$ is invertible.

Proof. Consider $\ker(A + I_n)$.

Let $\mathbf{v} \in \ker(A + I_n)$. Then

$$\begin{aligned}(A + I_n)\mathbf{v} &= \mathbf{0} \\ A\mathbf{v} + \mathbf{v} &= \mathbf{0} \\ A\mathbf{v} &= -\mathbf{v}.\end{aligned}$$

Multiplying A to the left on both sides gives

$$A^2\mathbf{v} = -A\mathbf{v} = (-1)(-\mathbf{v}) = (-1)^2\mathbf{v}.$$

Inductively, $\mathbf{0} = A^k\mathbf{v} = (-1)^k\mathbf{v} \implies \mathbf{v} = \mathbf{0}$. Hence, $\ker(A + I_n) = \{\mathbf{0}\}$.

By TFAE, since $A + I_n$ is a square matrix, $A + I_n$ is invertible. □

6 Change of coordinate matrix

6.1 Definition

Definition 7.6.1. Let V be a finite-dimensional vector space. Suppose S_1 and S_2 are two bases of V . Then the *change of coordinate matrix*, or *transition matrix*, from S_1 -basis to S_2 -basis is the matrix $[\text{id}_V]_{S_1}^{S_2}$.

Remark. Let $Q = [\text{id}_V]_{S_1}^{S_2}$.

1. Since id_V is an isomorphism, the matrix Q is invertible. Moreover,

$$Q^{-1} = [\text{id}_V]_{S_2}^{S_1}.$$

2. For any $\mathbf{v} \in V$, we have

$$[\text{id}_V]_{S_1}^{S_2}[\mathbf{v}]_{S_1} = [\mathbf{v}]_{S_2}.$$

In other words, the matrix $[\text{id}_V]_{S_1}^{S_2}$ converts the coordinate vector of $\mathbf{v}$ in the S_1 -basis into the coordinate vector of $\mathbf{v}$ in the S_2 -basis. This explains the name “change of coordinate” matrix. □

Definition 7.6.2. Let V be a vector space. A linear transformation $T : V \rightarrow V$ is called a *linear operator on V* or an *endomorphism on V* .

6.2 Theorem

Theorem 7.6.1. Let V be finite-dimensional and let $T : V \rightarrow V$ be a linear operator. Suppose

- S_1, S_2 are two bases of the vector space V ;
- $Q = [\text{id}_V]_{S_1}^{S_2}$ is the change of coordinate matrix from S_1 to S_2 .

Then we have an equality of matrices:

$$\boxed{[T]_{S_1}^{S_1} = Q^{-1}[T]_{S_2}^{S_2}Q.}$$

Proof. Consider the following diagram:

$$\begin{array}{ccc} V_{S_1} & \xrightarrow{[T]_{S_1}^{S_1}} & V_{S_1} \\ Q=[\text{id}_V]_{S_1}^{S_2} \downarrow & & \uparrow [\text{id}_V]_{S_2}^{S_1}=Q^{-1} \\ V_{S_2} & \xrightarrow{[T]_{S_2}^{S_2}} & V_{S_2} \end{array}$$

Here the subscripts indicate the bases used for the coordinates. Since

$$Q = [\text{id}_V]_{S_1}^{S_2} \quad \text{and} \quad [\text{id}_V]_{S_2}^{S_1} = Q^{-1},$$

$$\text{So } [T]_{S_1}^{S_1} = Q^{-1}[T]_{S_2}^{S_2}Q.$$

□

6.3 Example

6.3.1 Change of coordinate matrices

Let B be the standard basis of $\mathbb{R}^2$ and be the basis $S = \{(1, 3), (2, 5)\}$ of $\mathbb{R}^2$.

- Find the change of coordinate matrix from S to B .
- Find the change of coordinate matrix from B to S .
- Find $[(-2, 3)]_S$.

Solution. Let $B = \{\mathbf{e}_1, \mathbf{e}_2\}$.

(a) We have

$$\begin{aligned}\text{id}(\mathbf{1}, \mathbf{3}) &= (1, 3) = \mathbf{e}_1 + 3\mathbf{e}_2, \\ \text{id}(\mathbf{2}, \mathbf{5}) &= (2, 5) = 2\mathbf{e}_1 + 5\mathbf{e}_2.\end{aligned}$$

Therefore, $[\text{id}]_{\mathbf{S}}^B = \begin{pmatrix} 1 & 2 \\ 3 & 5 \end{pmatrix}$.

(b) By the inverse relation between change of coordinate matrices,

$$[\text{id}]_B^{\mathbf{S}} = ([\text{id}]_{\mathbf{S}}^B)^{-1} = \begin{pmatrix} 1 & 2 \\ 3 & 5 \end{pmatrix}^{-1} = \begin{pmatrix} -5 & 2 \\ 3 & -1 \end{pmatrix}.$$

(c) Since $[(-2, 3)]_B = \begin{pmatrix} -2 \\ 3 \end{pmatrix}$, we have

$$[(-2, 3)]_{\mathbf{S}} = [\text{id}]_B^{\mathbf{S}} [(-2, 3)]_B = \begin{pmatrix} -5 & 2 \\ 3 & -1 \end{pmatrix} \begin{pmatrix} -2 \\ 3 \end{pmatrix} = \begin{pmatrix} 16 \\ -9 \end{pmatrix}.$$

Indeed,

$$16(\mathbf{1}, \mathbf{3}) - 9(\mathbf{2}, \mathbf{5}) = (-2, 3).$$

□

6.3.2 Matrix Representations under Different Bases

Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the linear operator on $\mathbb{R}^2$ defined by reflection by the line $y = -x$. Let

$$B = \{(1, 0), (0, 1)\} \quad \text{and} \quad S = \{(1, 1), (1, -1)\}$$

be two bases of $\mathbb{R}^2$.

- (a) Find the matrix $X = [T]_B^B$.
- (b) Find the matrix $Y = [T]_S^S$.
- (c) Find the change of coordinate matrix Q from S to B .
- (d) Write down the relation between the matrices X, Y, Q .

Solution. (a)

$$T(1, 0) = (0, -1) = 0\mathbf{e}_1 - \mathbf{e}_2,$$

$$T(0, 1) = (-1, 0) = -\mathbf{e}_1 + 0\mathbf{e}_2.$$

$$\text{Therefore, } X = [T]_B^B = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}.$$

(b)

$$T((1, 1)) = (-1, -1) = -1(1, 1) + 0(1, -1),$$

$$T((1, -1)) = (1, -1) = 0(1, 1) + 1(1, -1).$$

$$\text{Therefore, } Y = [T]_S^S = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}.$$

(c)

$$\text{id}((1, 1)) = (1, 1) = \mathbf{e}_1 + \mathbf{e}_2,$$

$$\text{id}((1, -1)) = (1, -1) = \mathbf{e}_1 - \mathbf{e}_2.$$

$$\text{Hence, } Q = [\text{id}]_S^B = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

(d) Consider the following diagram:

$$\begin{array}{ccc} \mathbb{R}^2_{S\text{-basis}} & \xrightarrow{Y=[T]_S^S} & \mathbb{R}^2_{S\text{-basis}} \\ Q=[\text{id}]_S^B \downarrow & & \uparrow [\text{id}]_B^S = Q^{-1} \\ \mathbb{R}^2_{B\text{-basis}} & \xrightarrow{X=[T]_B^B} & \mathbb{R}^2_{B\text{-basis}} \end{array}$$

Since

$$X = [T]_B^B, \quad Y = [T]_S^S,$$

following the diagram gives

$$\boxed{Y = Q^{-1}XQ}.$$

Equivalently,

$$\boxed{X = QYQ^{-1}}.$$

□